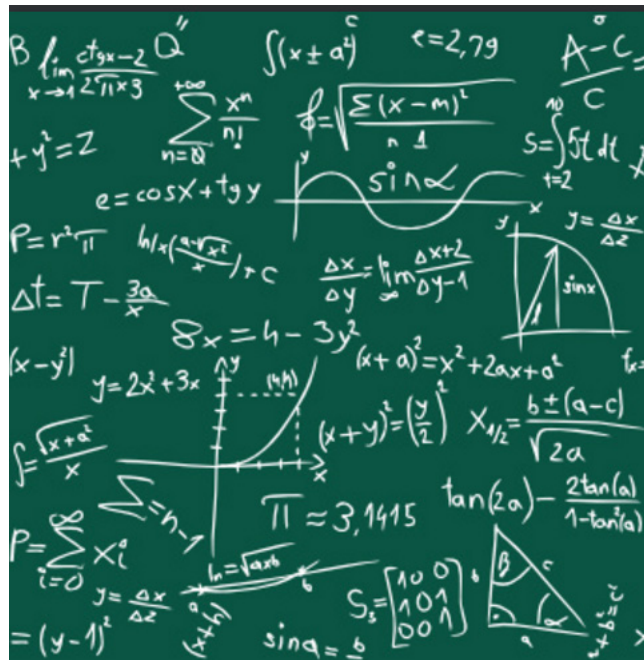


# Continuous functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

## Contents

|   |                                            |   |
|---|--------------------------------------------|---|
| 1 | Algebra of continuous functions theorem    | 2 |
| 2 | Continuity of elementary functions theorem | 2 |
| 3 | Continuity of composite functions theorem  | 5 |

# 1 Algebra of continuous functions theorem

## Theorem 1: Algebra of continuous functions

Let  $f$  and  $g$  be two functions defined at least in a neighborhood of  $x_0 \in \mathbb{R}$  and continuous at  $x_0$ . Then:

1.  $f(x) \pm g(x)$  is continuous at  $x_0$ ;      2.  $f(x) \cdot g(x)$  is continuous at  $x_0$ ;

3.  $\frac{f(x)}{g(x)}$  is continuous at  $x_0$  provided that  $g(x_0) \neq 0$ .

*Proof.* Let us prove, for example, claim 3., the others being analogous. By hypothesis we know that  $f$  and  $g$  are continuous at  $x_0$ , that is:

$$f(x) \rightarrow f(x_0) \text{ and } g(x) \rightarrow g(x_0) \text{ as } x \rightarrow x_0.$$

Moreover  $g(x_0) \neq 0$  and hence, by the sign-preservation theorem for continuous functions,  $g(x) \neq 0$  eventually as  $x \rightarrow x_0$ .

Then, by the theorem on the algebra of limits, we conclude that

$$\frac{f(x)}{g(x)} \rightarrow \frac{f(x_0)}{g(x_0)} \text{ as } x \rightarrow x_0$$

that is,  $f(x)/g(x)$  is continuous at  $x_0$ . □

# 2 Continuity of elementary functions theorem

## Theorem 2: Continuity of elementary functions

The following elementary functions are continuous at all points of their domain:

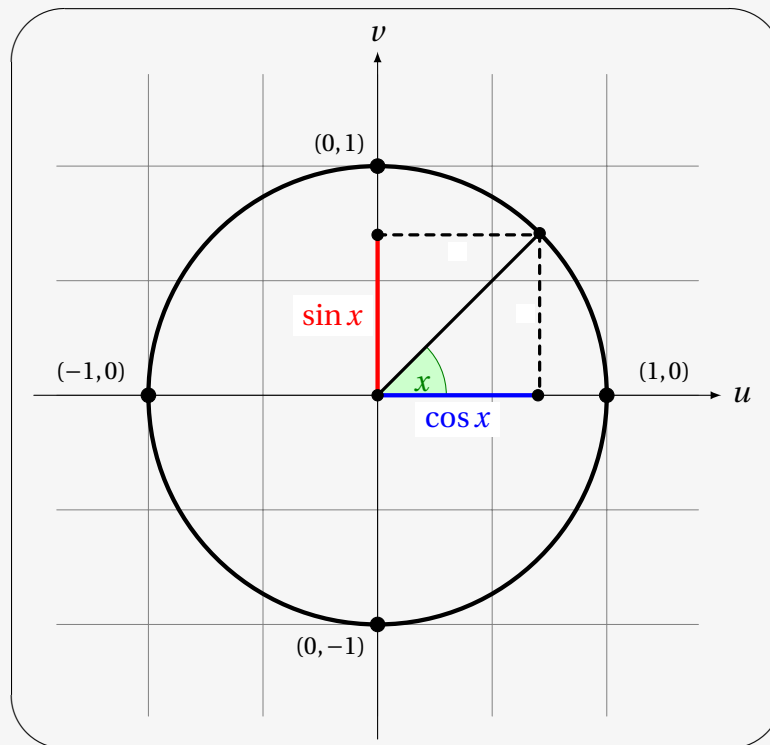
1. Powers with integer, rational or real exponent;
2. Exponential functions;
3. Logarithmic functions;
4. Elementary trigonometric functions ( $\sin x$ ,  $\cos x$ )

*Proof.* For example, let us prove the continuity on the whole of  $\mathbb{R}$  of the functions  $\sin x$  and  $\cos x$ .

- We have seen that  $\sin x$  is continuous at  $x = 0$ ; let us show that  $\cos x$  is also continuous at  $x = 0$ .
- The unit (trigonometric) circle shows that, if  $x$  is an angle in the first quadrant,

$$\sin^2 x + \cos^2 x = 1$$

since 1 is the hypotenuse of a right triangle with legs  $\sin x$ ,  $\cos x$ .



- It follows that

$$0 \leq 1 - \cos x \leq \sin x \quad \text{for } x \in \left[0, \frac{\pi}{2}\right], \text{ and hence}$$

$$0 \leq 1 - \cos x \leq |\sin x| \quad \text{for } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

- Then, by the comparison theorem,

$$1 - \cos x \rightarrow 0 \text{ as } x \rightarrow 0 \text{ hence } \cos x \rightarrow 1 \text{ as } x \rightarrow 0$$

and therefore  $\cos x$  is continuous at 0.

□

*Proof.* • To now prove the continuity of  $\sin x$  at a generic point  $x_0 \in \mathbb{R}$  we consider the chain of relations:

$$\begin{aligned} |\sin(x_0 + h) - \sin x_0| &= |\sin x_0 \cos h + \sin h \cos x_0 - \sin x_0| \\ &= |\sin x_0 (\cos h - 1) + \cos x_0 \sin h| \\ &\leq |\sin x_0 (\cos h - 1)| + |\cos x_0 \sin h| \\ &= |\sin x_0| |\cos h - 1| + |\cos x_0| |\sin h| \end{aligned}$$

Now  $|\sin h|$  and  $|\cos h - 1|$  tend to zero as  $h \rightarrow 0$ , by what we have just proved, while  $|\sin x_0|$  and  $|\cos x_0|$  are constants, so

$$\sin(x_0 + h) - \sin x_0 \rightarrow 0 \text{ as } h \rightarrow 0$$

that is

$$\sin(x_0 + h) \rightarrow \sin x_0 \text{ as } h \rightarrow 0$$

and  $\sin x$  is continuous at  $x_0$ . An analogous argument shows the continuity of  $\cos x$ .

□

- Having proved that  $\sin x$ ,  $\cos x$  are continuous on the whole of  $\mathbb{R}$ , we deduce that the functions  $\tan x$ ,  $\cot x$  are continuous on their domain (by the theorem on the algebra of continuous functions)
- Power functions with integer exponent are continuous (on the whole of  $\mathbb{R}$ ), since  $f(x) = x$  is obviously continuous, and  $f(x) = x^n$  is the product of  $n$  continuous functions.
- Polynomials are continuous functions, since they are obtained by adding functions of the type  $c x^n$ , which are continuous by the previous point.
- Rational functions (i.e., quotients of polynomials) are continuous functions, except at the points where the denominator vanishes (the denominator is a polynomial, hence it vanishes at a finite number of points)

### 3 Continuity of composite functions theorem

#### Theorem 3: Continuity of the composite function

Let:

- $g$  be a function defined at least in a neighborhood of  $x_0$  and continuous at  $x_0$ ,
- $f$  be a function defined at least in a neighborhood of  $t_0 = g(x_0)$  and continuous at  $t_0$ ,

then  $f \circ g$  is defined at least in a neighborhood of  $x_0$  and is continuous at  $x_0$ .

*Proof.* Since  $g$  is continuous at  $x_0$ ,

$$\lim_{x \rightarrow x_0} g(x) = g(x_0) = t_0$$

then, by the change of variable theorem for limits, we have:

$$\lim_{x \rightarrow x_0} f(g(x)) = \lim_{t \rightarrow t_0} f(t)$$

and since  $f$  is continuous at  $t_0$ , we have

$$\lim_{t \rightarrow t_0} f(t) = f(t_0)$$

and the thesis is proved. □

- It follows that all the functions that can be obtained from elementary functions by sums, products, quotients and compositions are continuous on their domain. **Hence, by combining functions in this way, we still obtain continuous functions.**

To summarize:

- sums, products and quotients of continuous functions give continuous functions (where the denominator does not vanish);
- the elementary functions of mathematical analysis are continuous on their domain;
- the composition of continuous functions gives a continuous function.

- It is therefore possible to know a priori that a function is continuous on its domain, without applying the definition of continuity case by case.

**For continuous functions**, if  $x_0$  is a point of the domain, the limit as  $x \rightarrow x_0$  is computed simply by substituting  $x_0$  into the analytic expression of the function, that is:

$$f(x) \rightarrow f(x_0) \quad \text{as } x \rightarrow x_0$$

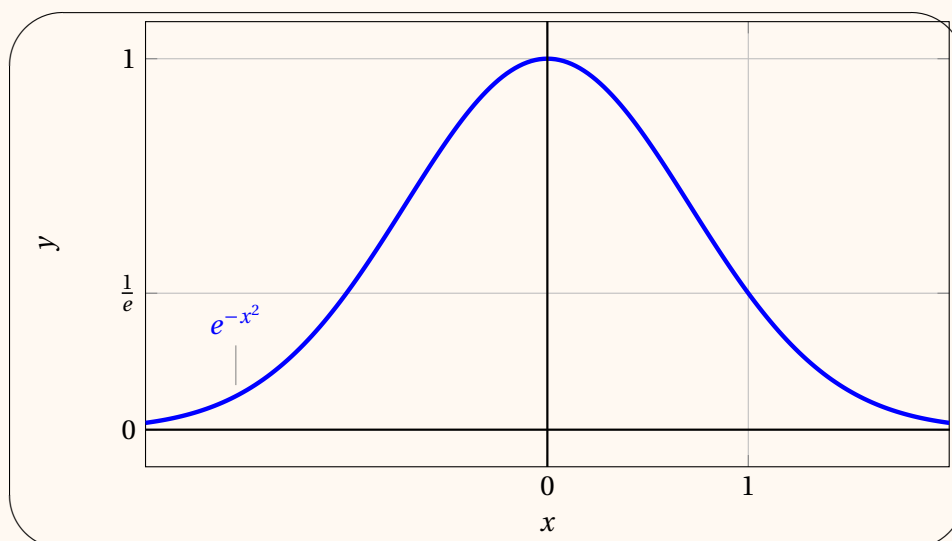
### Example 1: Continuous functions and finite limits at a finite point

The following function is continuous for  $x \in \mathbb{R}$ :

$$f(x) = e^{-x^2}$$

hence:

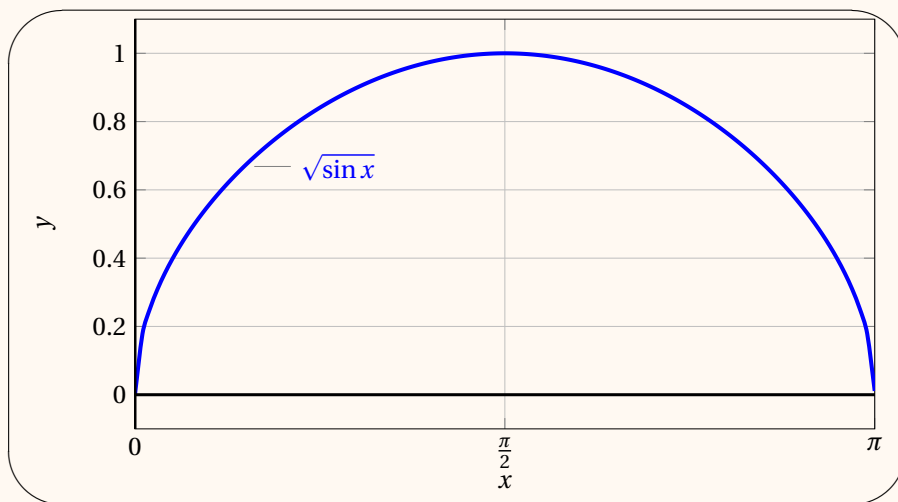
$$\lim_{x \rightarrow 1} e^{-x^2} = e^{-1} = \frac{1}{e}$$



### Example 2: Continuous functions and finite limits at a finite point

The following function is continuous for  $x \in \mathbb{R}$  with  $2k\pi \leq x \leq 2k\pi + \pi, \forall k \in \mathbb{Z}$ :

$$f(x) = \sqrt{\sin x} \quad \text{hence} \quad \lim_{x \rightarrow \pi^-} \sqrt{\sin x} = \sqrt{\sin \pi} = 0$$



### Example 3: Continuous functions and finite limits at a finite point

The following function is continuous for  $x \in \mathbb{R}$  with  $x \neq (2k+1)\frac{\pi}{2}, \forall k \in \mathbb{Z}$ :

$$f(x) = \log_a(1 + (\tan x)^2), \quad \forall a > 0, a \neq 1$$

hence:

$$\lim_{x \rightarrow 0} \log_a(1 + (\tan x)^2) = \log_a(1 + (\tan 0)^2) = \log_a 1 = 0$$

For example, with  $a = e$  we have:

